RAJALAKSHMI ENGINEERING COLLEGE

Abirami G 251001004-2025-IT

A2

Started on	Tuesday, 25 August 2026, 11:26 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:27 AM
Time taken	44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, c = 0;
    int i, j, k;

    scanf("%d", &n);

    c++;   // original c++

    for (i = n / 2; i < n; i++)
    {
        c++;   // outer condition

        for (j = 1; j < n; j = j * 2)
        {
            c++;   // middle condition

            for (k = 1; k < n; k = k * 2)
            {
                c++;   // inner condition
                c++;   // original c++
            }

            c++;   // final inner condition
        }

        c++;   // final middle condition
    }

    c++;   // final outer condition

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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